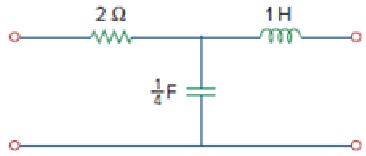
Chapter 19, Problem 50P

Problem

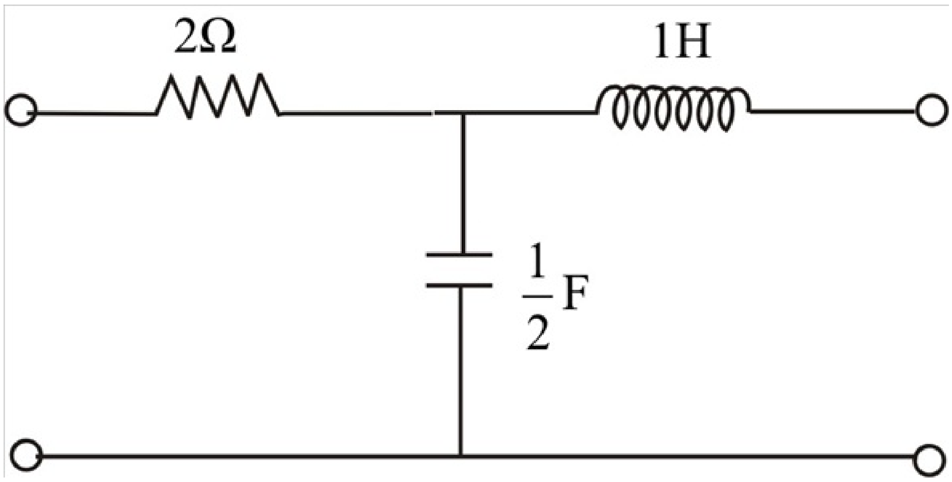
Derive the *s*-domain expression for the *t* parameters of the circuit in Fig. 19.104.

Figure 19.104

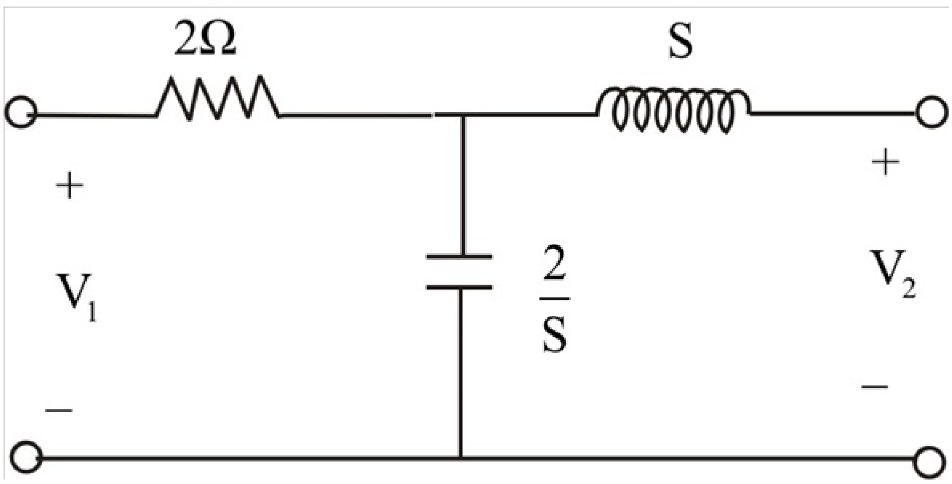


Step-by-step solution

Step 1 of 7



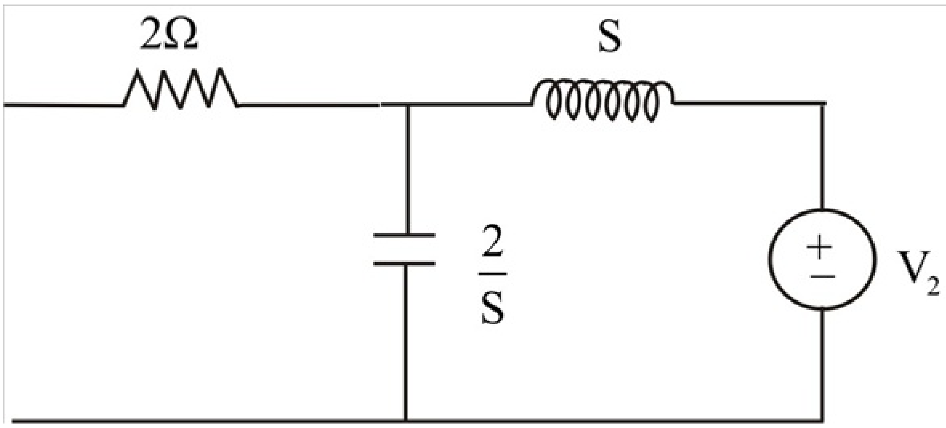
Step 2 of 7



Step 3 of 7

$a = \left.\frac{V_2}{V_1}\right|_{I_1=0}$ $\quad b = \left.\frac{-V_2}{I_1}\right|_{V_1=0}$ $\quad c = \left.\frac{I_2}{V_1}\right|_{I_1=0}$ $\quad d = \left.\frac{-I_2}{I_1}\right|_{V_1=0}$

From $I_1 = 0$



Step 4 of 7

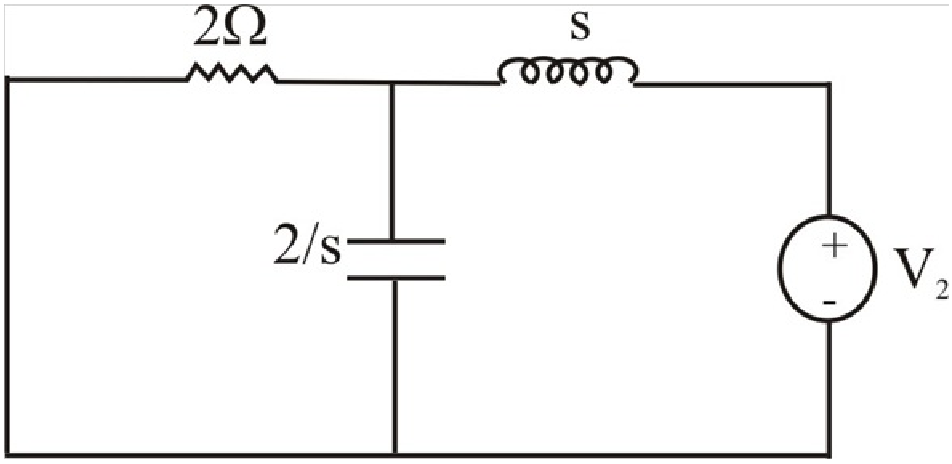
$$\frac{V_a}{V_1} = \frac{V_1 \frac{1}{\sqrt{\left(S+\frac{2}{S}\right)\times\frac{2}{S}}}}{V_1}$$

$$= \frac{\frac{\frac{1}{2}}{S}}{\frac{\left(S^2+2\right)}{S}} = \frac{\frac{1}{2}}{S^2+2} = \frac{S^2+2}{2}$$

$$C = \frac{I_2}{V_1} = \frac{\frac{V_2}{S+\frac{2}{S}}}{\frac{V_2}{S+\frac{2}{S}}\times\frac{2}{S}} = \frac{S}{2}$$

Step 5 of 7

For $V_1 = 0$



Step 6 of 7

$$\frac{-V_2}{I_1} = \frac{-V_2}{\left(I_2 + \frac{2}{S}\right)\left(\frac{2}{S} + 2\right)}$$

$$= \frac{-I_2\left(2 \,||\, \frac{2}{S} + s\right)}{\frac{I_2 \frac{2}{S}}{\frac{(2S+2)}{S}}}$$

$$= \frac{\frac{-\frac{4}{S}}{2S + \frac{2}{S}} + S}{\frac{1}{(2S+2)}}$$

$$= -4 + 2S^2 + 2S = -2\left(S^2 + S + 2\right)$$

Step 7 of 7

$$\frac{-I_2}{I_1} = \frac{-I_2}{I_2\left(\frac{2}{S}\right) \times \frac{1}{2 + \frac{2}{S}}} = \frac{-1}{\frac{2}{S}\left(\frac{1}{\frac{2S+2}{S}}\right)} = -(S+1)$$

$$\boxed{a = \frac{S^2+2}{2} \qquad b = -2\left(S^2 + S + 2\right) \qquad c = \frac{S}{2} \qquad a = (-S+1)}$$